

Functional Analysis — Homework 5

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Problem 1. (Spectrum of a product). Find an example of two linear operators $A, B : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\sigma(A) = \sigma(B) = \{0\}$, but $\sigma(AB) \neq \{0\}$. Thus, the spectrum of the product AB can be strictly larger than the spectra of A and B together.

Proof. Consider

$$\begin{aligned} A &= \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \\ B &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\ AB &= \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

using the standard basis. Then for $\lambda \in \mathbb{R}$, we find solutions for

$$\begin{aligned} \det(A - \lambda) &= \det \begin{pmatrix} -\lambda & -1 \\ 0 & -\lambda \end{pmatrix} = \lambda^2 = 0 \\ \det(B - \lambda) &= \det \begin{pmatrix} -\lambda & 0 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 = 0 \\ \det(AB - \lambda) &= \det \begin{pmatrix} -1 - \lambda & 0 \\ 0 & -\lambda \end{pmatrix} = \lambda(1 + \lambda) = 0. \end{aligned}$$

The first two only have the solution $\lambda = 0$, whereas the third has solutions at $\lambda = 0$ and $\lambda = -1$. Thus $\sigma(A) = \sigma(B) = \{0\}$ and $\sigma(AB) = \{0, -1\}$ as desired. $\square$

Problem 2. Let $[a, b] \subseteq [0, 1]$ with $a < b$. Find a bounded linear operator $T : C^0[0, 1] \rightarrow C^0[0, 1]$ such that $\sigma(T) = [a, b]$.

Proof. Define $T(f)(t) := (a + t(b - a))f(t)$. This is linear since

$$\begin{aligned} T(\alpha f + \beta g)(t) &= (a + t(b - a))(\alpha f + \beta g)(t) \\ &= (a + t(b - a))(\alpha f(t) + \beta g(t)) \\ &= \alpha(a + t(b - a))f(t) + \beta(a + t(b - a))g(t) \\ &= \alpha T(f)(t) + \beta T(g)(t) \\ &= (\alpha T(f) + \beta T(g))(t) \end{aligned}$$

and bounded because

$$\begin{aligned} \|T(f)(t)\| &= \|(a + t(b - a))f(t)\| \\ &= (a + t(b - a))\|f(t)\| \\ &\leq b\|f(t)\|. \end{aligned}$$

for all $t \in [0, 1]$. Now if $\lambda \notin [a, b]$, then $(T - \lambda)(f)(t) = g(t)$ can be expanded to

$$(a + t(b - a) - \lambda)f(t) = g(t).$$

Since $a + t(b - a) \in [a, b]$ for all $t \in [0, 1]$ and $\lambda \notin [a, b]$, we have $(a + t(b - a) - \lambda) \neq 0$ for all $t \in [0, 1]$. Thus we can solve the equation with $f(t) = g(t)/(a + t(b - a) - \lambda)$, so T is surjective and $\lambda \in \rho(T)$.

Now if $\lambda \in [a, b]$, then

$$\text{im}(T - \lambda) \subseteq \left\{ f \in C^0[0, 1] : f\left(\frac{\lambda - a}{b - a}\right) = 0 \right\}.$$

This cannot be dense in $C^0[0, 1]$, so $\lambda \in \sigma_r(T)$, and we have shown that $\sigma(T) = [a, b]$. $\square$

Problem 3. For a bounded linear operator $T : X \rightarrow X$, show that

$$\|(T - \lambda)^{-1}\|_{op} \rightarrow 0 \text{ as } \lambda \rightarrow \infty.$$

Here, $\lambda \rightarrow \infty$ should be understood as “for $|\lambda|$ sufficiently large”, $\lambda \in \mathbb{C}$. We need only consider λ outside the closed ball $B_{\|T\|_{op}}[0]$, where $(T - \lambda)$ is invertible.

Proof. Note that since $(1 - T)^{-1} = 1 + T + T^2 + \dots$, if we choose λ large enough that $|\lambda| \geq 2\|T\|_{op}$, so that $\|T/\lambda\|_{op} \leq 1/2$, we have

$$\begin{aligned} \|(T - \lambda)^{-1}\|_{op} &= \|(T/\lambda - 1)^{-1}\|_{op} \\ &= \|(1 - T/\lambda)^{-1}\|_{op} \\ &= \|(1 - T/\lambda)^{-1}\|_{op} \\ &= \left\| 1 + \frac{T}{\lambda} + \left(\frac{T}{\lambda}\right)^2 + \dots \right\|_{op} \\ &\leq \|1\|_{op} + \left\| \frac{T}{\lambda} \right\|_{op} + \left\| \left(\frac{T}{\lambda}\right)^2 \right\|_{op} + \dots \\ &\leq 1 + \left\| \frac{T}{\lambda} \right\|_{op} + \left\| \left(\frac{T}{\lambda}\right)^2 \right\|_{op} + \dots \\ &\leq 1 + \frac{1}{2} + \frac{1}{4} + \dots \\ &= 2. \end{aligned}$$

Choosing any integer n instead of 2 gives us a smaller upper bound, but with 1 as the lowest upper bound possible with this method, due to the fact that $\|1\|_{op} = 1$. I’m not sure how to get around that yet. $\square$

Problem 4. For a bounded linear operator $T : X \rightarrow X$ and $\lambda \in \rho(T)$ (resolvent set), write $R_\lambda = (T - \lambda)^{-1}$. Prove the resolvent equation:

$$R_\lambda - R_\mu = (\lambda - \mu)R_\lambda R_\mu, \quad \lambda, \mu \in \rho(T).$$

Proof. Denoting the left-hand side of the resolvent equation as LHS and the right-hand side as RHS, we will prove that

$$(T - \mu)(T - \lambda)(LHS) = (T - \mu)(T - \lambda)(RHS).$$

Since $(T - \mu)(T - \lambda)$ itself has an inverse, this will show that $LHS = RHS$.

First, the RHS:

$$\begin{aligned} (T - \mu)(T - \lambda)(RHS(x)) &= (T - \mu)(T - \lambda)(\lambda - \mu)(T - \lambda)^{-1}(T - \mu)^{-1}(x) \\ &= (\lambda - \mu)(T - \mu)(T - \lambda)(T - \lambda)^{-1}(T - \mu)^{-1}(x) \\ &= (\lambda - \mu)x \\ &= \lambda x - \mu x. \end{aligned}$$

For the LHS, we derive the following strange identity

$$\begin{aligned} (T - \mu)(T - \mu)^{-1}(x) &= x \\ \implies T(T - \mu)^{-1}(x) - \mu(T - \mu)^{-1}(x) &= x \\ \implies T(T - \mu)^{-1}(x) &= \mu(T - \mu)^{-1}(x) + x. \end{aligned}$$

Now we have

$$\begin{aligned} (T - \mu)(T - \lambda)(LHS(x)) &= (T - \mu)(T - \lambda)((R_\lambda - R_\mu)(x)) \\ &= (T - \mu)(T - \lambda)((T - \lambda)^{-1} - (T - \mu)^{-1})(x) \\ &= (T - \mu)(T - \lambda)(T - \lambda)^{-1}(x) - (T - \mu)^{-1}(T - \lambda)(T - \mu)(x) \\ &= (T - \mu)(x) - (T - \mu)(T - \lambda)(T - \mu)^{-1}(x) \\ &= T(x) - \mu x - (T - \mu)(T(T - \mu)^{-1}(x) - \lambda(T - \mu)^{-1}(x)) \\ &= T(x) - \mu x - ((T - \mu)(T(T - \mu)^{-1}(x)) - \lambda x) \\ &= T(x) - \mu x - (T - \mu)(\mu(T - \mu)^{-1}(x) + x) + \lambda x \\ &= T(x) - \mu x - (\mu x + (T - \mu)(x)) + \lambda x \\ &= T(x) - \mu x - T(x) + \lambda x \\ &= \lambda x - \mu x, \end{aligned}$$

so we are done. (Interested to see if you have a more enlightening way of doing this.) $\square$